Aurix TC275SB: Virtual Flight Controller

Hardware

Aurix TC275SB with **ClickShield (CS),** 13DOF Click/6DOF IMU 14 Click/BMI088 I2C Click/BMI088 SPI Click, OLED C Click

Goal

Use the given hardware to create a virtual quadcopter flight controller.

Function Description

Use the Project template where the UART (via USB) is already implemented.

Use all three cores of the Aurix board:

- **Core 0** uses the IMU ClickBoard to measure the board's orientation. It communicates with the IMU ClickBoard via I^2C or SPI and sends forth the measured data that has been scaled to physical units (g, °/sec) via shared memory.
- **Core 1** receives this data and uses them to estimate Roll/Pitch/Yaw angles (see, e.g., [1]). It sends these angles forth via shared memory.
- **Core 2** receives these angles and sends them out via UART in a format compatible with the IMU visualizer available in [2]. Furthermore, it implements a simple P controller [3] that determines the RPM of the four motors to bring a hypothetical quadcopter into level flight based on [4]. Finally, it displays the RPMs shall be on an OLED C click with which it communicates via SPI. You may use template [5] as the OLED background.

Core 0 and Core 1 can exchange data via a shared memory (SM) as can Core 0 and Core 2. (There are two shared memories, one for Core 0 and 1 and one for Core 0 and 2). The shared memories **must** be protected. The core that uses the shared memory must claim ownership before reading or writing the shared memory and release it. Use a **mutex** to protect the memories.

Relevant File

Custom UART files:

- UART.h
- UART.c

  Use these files to expand UART functionality and add custom wrapper functions for functions defined in IfxAsclin_Asc.h and IfxAsclin_Asc.c

Aurix UART files:

- Asclin/Asc/IfxAsclin_Asc.h
- Asclin/Asc/IfxAsclin_Asc.c
  UART functionality provided by Aurix. Used in the custom UART Files

Aurix Interrupt files:

- Cpu/Irq/IfxCPU_Irq.h
- Cpu/Irq/IfxCPU_Irq.c
  Interrupt files (needed for UART communication)

Aurix Mutex files:

- Cpu/Std/IfxCpu.h
- Cpu/Std/IfxCpu.c
  Mutex functionality provided by the Aurix board

Aurix SPI Files are in the Folder
`Libraries/ILLD/TC27D/QSPI`

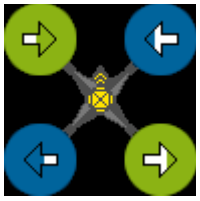
Aurix I2C Files are in the Folder
`Libraries/ILLD/TC27D/I2C`

Hint: You can find the Aurix files in the Project under: `Libraries/iLLD/TC27D/Tricore/<PATH_TO_FILE>`
i.e.: `Libraries/iLLD/TC27D/Tricore/Asclin/Asc/IfxAsclin_Asc.h`

Hardware

If you use an I2C IMU, use CON 1 for I^2C and CON 2 for SPI communication.
Note that problems have been reported concerning SPI communication via CON 1.

Virtual Flight Controller



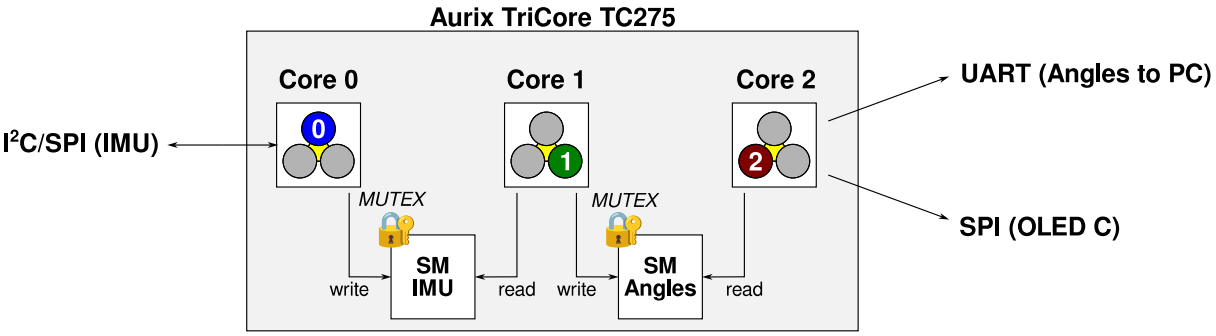
Example showing M1=75%, M2=50%, M3=50%, M4=75% [2]:

Relevant Documents

- Aurix_Unleashed.pdf
- TC275SB_clickShield_PinLayout.pdf
- TC275SB_clickShield_Schematic.pdf

Implementation Hints

[1] https://ahrs.readthedocs.io/en/latest/filters/complementary.html
[2] https://embsys.technikum-wien.at/projects/drohnfh/imu-visualizer/imu-visualizer.php
[3] https://en.wikipedia.org/wiki/Proportional_control
[4] https://cookierobotics.com/066/
[5] See Moodle for Template Image .h/.ppm/.png and image conversion script.



Acknowledgment

Stadt Wien Kompetenzteam für Drohnentechnik
in der Fachhochschulausbildung
MA23 Projekt 35-02